

SYSTEMS ANALYSIS

Systems Analysis & Design

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Outline

- 1 Basic Concepts
- 2 Chaos and Dynamic Systems
- 3 Abstraction and Modularity



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What is a System?

- A **system** is a **set** of **interacting components** that *work* together to **achieve** a **common goal**.
- A **system** is a **collection** of **elements organized** in a specific way.
- A **system** is a **structure designed** to **perform** a specific **function**.



Systems Analysis Process

- **Systems analysis** is the *process* of **studying** a **system** in order to identify its **components**, **interactions**, and **goals**.
- Systems analysis is the *process* of **understanding** how a **system** **works** and how it can be **improved**.
- Systems analysis is the *first* **step** in the **systems development** **lifecycle**.



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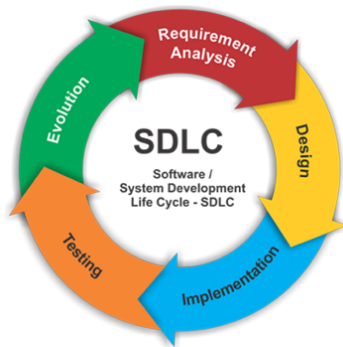
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Systems Development Lifecycle

- The **systems development lifecycle** is a *process* that *guides* the *development* of a *system*.
- It includes *planning*, *analysis*, *design*, *implementation*, and *maintenance* phases.
- It is a **structured approach** to *developing* a *system* that *ensures* it *meets* the *needs* of its *users*.



Systems Analysis Techniques

- **Systems analysis** uses a *variety* of **techniques** to *study* a **system**.
- These *include* **interviews**, **surveys**, **observations**, and **document analysis**.
- It also *includes* **data modeling**, **process modeling**, and **requirements analysis**.



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Systems Analysis Tools

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- These *include* **diagrams**, **charts**, **flowcharts**, and **data models**.
- It also *includes* software tools such as **spreadsheets**, **databases**, and **simulation software**.



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Lateral Thinking

- **Lateral thinking** is a **creative problem-solving technique** that involves **thinking outside** the **box**.
- It is a **non-linear** approach to **problem-solving** that *encourages* **innovation** and **creativity**.
- It is a *useful* **technique** for *generating* **new ideas** and **solving complex problems**.
- Examples:
 - How can you improve the design of a product?
 - What are the benefits of failure?
 - Why is ignorance important?
 - When is failure better than success?



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Lateral Thinking Training

- **Lateral thinking** is a *skill* that can be **learned** and **developed** through training and practice.
- It involves **exercises**, **games**, and **activities** that **encourage creative thinking**.
- Examples of lateral thinking exercises:
 - Brainstorming sessions.
 - Mind mapping exercises.
 - Role-playing games.
 - Problem-solving activities.



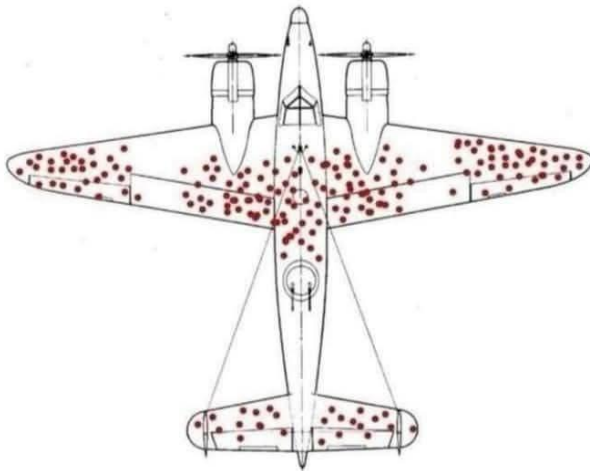
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Lateral Thinking Exercise

World War II, the perspective of **Abraham Wald**:



Uncertainty and Risk

- **Uncertainty** is the **lack** of **knowledge** about the **future outcome** of a decision or event.
- **Risk** is the **probability** of a **negative outcome** or **loss** associated with a decision or event.
- **Uncertainty** and **risk** are *inherent* in **complex systems** and **decisions**.
- They can be **managed** through **planning**, **analysis**, and **mitigation strategies**.



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Complexity and Emergence

- **Complexity** is the **degree** to which a **system** is **difficult** to **understand**.
- **Emergence** is the **appearance** of **unexpected** properties in a system that **arise** from the **interactions** of its **components**.
- **Complexity** and **emergence** are *common* in **dynamic** systems that are **non-linear** and **chaotic**.
- They can be studied and understood through **systems analysis** and **modeling**.



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What is Chaos?

- **Chaos** is a *branch* of **mathematics** that *studies* the **sensitivity** of **dynamical systems** to **initial conditions**.
- **Chaos** is a **non-linear** behavior that is **highly sensitive** to **initial conditions**.
- **Chaos** is a **deterministic** behavior that is **not predictable** in the **long term**.
- **Chaos** is a **complex** behavior that is **hard** to understand.

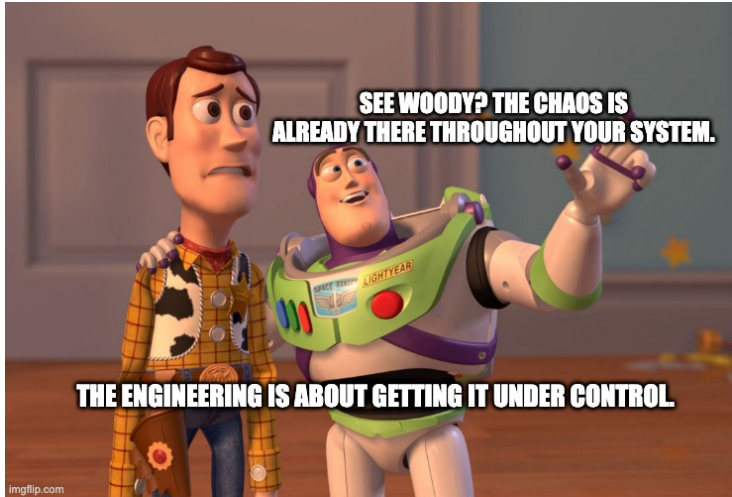


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Chaos is Everywhere!



What is a Dynamic System?

- A **dynamic system** is a system that **changes** over time.
- A **dynamic system** is **sensitive** to **initial conditions**.
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Chaotic Attractors

A **chaotic attractor** is a set of points in a phase space that attracts the trajectories of a dynamical system. One of the most famous chaotic attractors is the **Lorenz attractor**, discovered by *Edward Lorenz* in 1963 while studying a simplified model of atmospheric convection. It is defined by the following set of differential equations:

$$\frac{dx}{dt} = \sigma(y - x)$$

$$\frac{dy}{dt} = x(\rho - z) - y$$



Fractals in Nature



Watch this video: <https://www.youtube.com/watch?v=kkGeOWYOFoA>



Swarm Intelligence I

- **Swarm intelligence** is the collective behavior of *decentralized, self-organized systems*, natural or artificial.
- The concept is employed in work on **artificial intelligence**.
- The expression was introduced by *Gerardo Beni* and *Jing Wang* in 1989, in the context of **cellular robotic systems**. For example, watch this **video**.



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Swarm Intelligence II

- The **idea** is: if you observe an **individual**, it may seem random; **however**, several **individuals interacting** with each other and the environment show **intelligent behaviors**.
- There are interesting **population behaviors** in nature, especially in **insects**: bees, ants, termites, among others.
- There are also many **examples in nature**: schools of fish, birds, wolves.

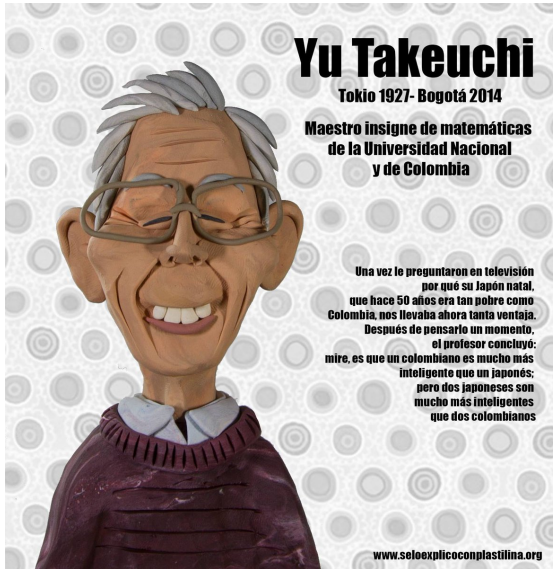


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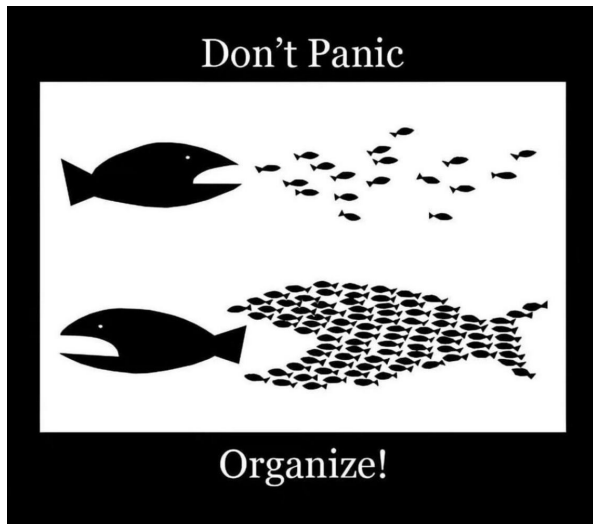


Emergent Behaviors

- **Emergent behavior** is the **appearance** of **complex patterns** and behaviors from **many** simple interactions.
- **Emergent behavior results** from the **collective** behavior of the **individuals** in the system.
- **Emergent behavior** is **not planned** or **designed** by any individual, but **arises** from their **interactions**.
- **Emergent behavior** is **not** the **sum** of the **individual** behaviors, but **something more**: **synergy**.
- **Swarm intelligence** refers to interesting **emergent behaviors**.



School Fish Algorithm



School Fish Algorithm

- **School fish** are interesting. When a predator attacks, they become confused by the large number of individuals and their **diverse movements**.
- The **idea** is simple: *"Don't touch me, don't come too close, but stay somewhat close."*
- This behavior creates a **chain of action and reaction**. It **confuses predators** and helps the school move uniformly.
- Do you remember **Nemo**? The fish with the sword snout, the pirates, or Marlins imitation of talking-all are somewhat similar. Watch **here**.
- The **school fish algorithm** is a **multi-agent system** that **simulates** the behavior of a **school of fish**.



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- The **ant colony algorithm** is based on the **social behavior** of **ants** and the use of **pheromones**. Watch **here**.
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- **Abstraction** is the *process* of **simplifying** a **complex system** in order to **understand** it better.
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- **Modularity** is the *process* of **organizing** a **system** into **independent units** that can be **developed** and **maintained** separately.
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Abstraction and Modularity

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Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/systems-analysis>

